



Kulong Lam Wuol Nyar
TC#257904

Software Engineer | Full-Stack Developer | Backend & API Development Specialist | Technical Support & Software Quality Assurance Professional

PROFILE SUMMARY

Software Engineer with hands-on experience in full-stack web application development, technical troubleshooting, database management, and software quality assurance. Skilled in designing scalable applications using Python, Django, Node.js, JavaScript, PostgreSQL, and MongoDB. Experienced in debugging, validating application functionality, **analyzing requirements**, and supporting end users in technical environments. Strong analytical, communication, and problem-solving skills with a passion for software testing, technical support, and delivering reliable customer-focused solutions.

PROFESSIONAL SKILLS

- **Technical Skills:** Python, JavaScript, Node.js, Django, Express.js, Vue.js, HTML, CSS, Bootstrap, Tailwind CSS, PostgreSQL, MongoDB, MySQL, Windows, Linux, Trouble shooting, Debugging, System design and architecture, Database Design, Agile Methodology.
- **Tools:** Git, GitHub, Vs code, Postman, Jira
- **Soft Skills:** Problem-solving, Customers support, Analytical thinking, Time management, Communication, Team Collaboration, Leadership

EDUCATION

- **Bachelor Degree – Software Engineering- Jul 2023-** Bugema University - Kampala, Uganda

WORK EXPERIENCE

Full-Stack Developer – Karibu Groceries Management System

Jan- 2026- present

Student Project – Node.js, Express.js, MongoDB, Vue.js

- Design and develop a full-stack grocery management system using Node.js, Express.js, MongoDB, and Vue.js.
- Develop and validate REST APIs supporting inventory, procurement, cash sales, and credit sales.
- Test application functionality, verify business logic, and resolve software defects throughout the development lifecycle.
- Perform database validation using MongoDB queries to ensure data integrity and application reliability.
- Implement JWT authentication and role-based authorization for secure user access.
- Deploy and maintain the application on Render Cloud.

IT Intern (Industrial Training)

June 2023 – August 2023

Web Info-net Ltd – Kampala, Uganda

- Installed and configured Windows and Linux operating systems.
- Installed and maintained LAN/WAN network infrastructure.
- Diagnosed and resolved hardware and software issues for end users.
- Installed and managed databases and web hosting services.
- Supported website deployment and maintenance.
- Provided technical support and troubleshooting to users.
- Assisted in server administration and computer maintenance.

Full-Stack Developer – Human Resource Management Information System

Jan-2023- Dec 2023

Final Year Project – Bugema University, Kampala, Uganda – Django, PostgreSQL, JavaScript

- Built and tested a full-stack HRMIS using Django REST Framework and React, covering payroll, attendance, leave, recruitment and reporting.
- Created **131 automated API tests**, Vitest unit tests and Playwright smoke tests for critical workflows.
- Diagnosed and resolved application, database migration, API, dependency and deployment issues.
- Implemented CI checks, health monitoring, request tracing, backup/restore procedures and operational runbooks.
- Applied MFA, RBAC, encryption, audit logging, upload validation and API rate limiting to protect sensitive HR data.
- Developed Uganda-specific PAYE/NSSF exports and payroll reconciliation tools.

PROFESSIONAL DEVELOPMENT

Trainings/Workshops:

- Certificate in Software Engineering Training (JavaScript, Node.js, MongoDB, Vue.js) – Refactory Academy – Kampala, Uganda – Pending Graduation.

Certificates:

- Google Data Analytics Professional Certificate (In Progress) – Coursera
- Foundation in Cybersecurity – Coursera
- International Computer Driving Licence (ICDL) – Engage Consults – 08/08/2025

PUBLICATIONS/PROJECTS

Projects:

- Karibu Groceries Management System (Node.js, Express.js, MongoDB, Vue.js) – <https://kgl-frontend.onrender.com>
- Human Resource Management Information System (Python, Django, PostgreSQL) – <https://github.com/kulonglam>

Publications:

- Human Resource Management Information System (Final Year Project) – Bugema University – Kampala, Uganda – 28/07/2023

LANGUAGE COMPETENCIES

- Nuer: Native
- English: C1/C2
- Italian: A1

HOBBIES

- Backend Engineering & API Development, Full-Stack Web Applications, Data-Driven Systems, Frontend Development